

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course : Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTES304 - ELECTRICAL MACHINES AND INSTRUMENTS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The main function of the commutator in a DC machine is to:
a) Reduce armature reaction b) Convert AC to DC
c) Reverse the direction of current in armature conductors d) Increase efficiency
- 2 In a DC motor, the back emf:
a) Increases the armature current b) Helps in starting
c) Opposes the applied voltage d) Acts as a braking force
- 3 A four-point starter is used for:
a) Series motors b) Compound motors c) Shunt motors d) All of the above
- 4 In an induction motor, the rotor never reaches synchronous speed because:
a) Of high rotor resistance b) Back emf is produced
c) Torque becomes zero d) Core losses are high
- 5 Which starter is suitable for a 3-phase squirrel cage induction motor?
a) DOL starter b) Star-delta starter
c) Auto-transformer starter d) All of the above
- 6 The device used to improve the power factor using a synchronous motor is:
a) Induction regulator b) V-curve controller
c) Synchronous condenser d) Capacitor bank
- 7 A stepper motor operates based on:
a) Electromagnetic induction b) Rotor inertia
c) Discrete steps d) Linear motion
- 8 A servo motor is mainly used in:
a) Welding b) Speed control systems c) Drilling d) Transformers
- 9 A strain gauge is used to measure:
a) Pressure b) Force c) Temperature d) Displacement
- 10 Which of the following transducers is sensitive to light?
a) Strain gauge b) LVDT c) Photodiode d) Thermocouple
- 11 Vibration in machines is measured using:
a) Strain gauge b) Tachometer c) Accelerometer d) LVDT

- 12 Smoke detectors work on the principle of:
 a) Thermocouple b) Radiation c) Optical sensing d) Induction

- Q2. Solve the following.
- A) What are the different methods of speed control of DC motors? 6
- B) Explain the necessity of a starter in DC motors. Describe the working of a three-point starter. 6
- Q3. Solve the following.
- A) Explain the construction and working principle of a three-phase induction motor. 6
- B) Explain V-curves of synchronous motors and their significance. 6
- Q4. Solve Any Two of the following.
- A) Describe the construction and working of a stepper motor. What are its types and applications? 6
- B) Discuss the construction, working, and applications of a hysteresis motor. 6
- C) Explain the working of a repulsion motor with a neat sketch. 6
- Q5. Solve Any Two of the following.
- A) Describe the working principle and applications of strain gauge. 6
- B) Write a note on Hall Effect transducers and their use in magnetic field measurement. 6
- C) Differentiate between analog and digital transducers with suitable examples. 6
- Q6. Solve Any Two of the following.
- A) What is electrical telemetry? Discuss its types and applications. 6
- B) Explain the principle and working of computerized tomography. 6
- C) Discuss the working and applications of an X-Y plotter. 6
- *** End ***